

HW7

P1. $I_C R_C = 10\text{k}\Omega \times 0.2\text{mA} = 2\text{V}$

$$V_C = V_{CC} - I_C R_C = 3\text{V} - 2\text{V} = 1\text{V}$$

$$R_E = \frac{V_I}{I} = \frac{25\text{mV}}{0.2\text{mA}} = 125\Omega$$

$$\frac{V_C}{V_I} \approx \frac{R}{R_E} = \frac{10\text{k}\Omega}{125\Omega} = 80$$

P2. $V_B = \frac{5}{2} = 2.5\text{V}$ $V_E = V_B - 0.7 = 1.8\text{V}$

$$I_E = \frac{1.8}{3.6\text{k}} = 0.5\text{mA}$$

$$r_e = \frac{25\text{mV}}{0.5\text{mA}} = 50\Omega$$

$$\frac{V_{O1}}{V_I} \approx \frac{R_E}{R_E + r_e} = \frac{3600}{3600 + 50} = 0.986$$

$$\frac{V_{O2}}{V_I} \approx \frac{-3300}{3600 + 50} \approx -0.904$$

$$\frac{V_{O2}}{V_I} \approx -\frac{R_C}{r_c} = -\frac{3300}{50} \approx -66$$